High School Life Skills Mathematics Lesson Plan

Lesson Information

Student Teacher: _____
Cooperating Teacher: _____
School: _____
Date: _____
Grade/Program: High School Life Skills Mathematics
Subject: Mathematics / Functional Academics
Lesson Length: 55 minutes
Lesson Topic: Making Purchases and Calculating Change

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1. Pennsylvania Standards and Competencies

Pennsylvania Core Mathematics Standards

This lesson connects to the Pennsylvania Core Mathematics Standards and Standards of Mathematical Practice, particularly the expectation that students apply mathematical concepts to real-world situations.

PA Core Mathematics – Numbers and Operations:
Students will apply addition and subtraction skills to solve real-world problems involving quantities, including money.

Standards for Mathematical Practice:

- **MP1 – Make sense of problems and persevere in solving them.**
- **MP2 – Reason abstractly and quantitatively.**
- **MP4 – Model with mathematics.**
- **MP5 – Use appropriate tools strategically.**
- **MP6 – Attend to precision.**

Pennsylvania identifies these mathematical practices as an integral part of its mathematics standards and emphasizes the application of mathematics to real-world issues and challenges.

Pennsylvania Personal Finance Connection

The lesson also supports Pennsylvania's current Personal Finance framework, particularly the area of **Spending**. Pennsylvania's Personal Finance standards organize financial literacy into Personal Finance Fundamentals, Income, Spending, Saving & Investing, Risk & Insurance, and Credit.

Functional Competency:
Students will demonstrate the ability to use money to make a purchase, calculate the total cost of items, and determine the appropriate amount of change.

Functional Life Skills Competency

Students will apply mathematical skills to a simulated shopping situation in order to increase independence with everyday financial tasks.

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2. Big Idea

Mathematics can be used to solve practical problems that students encounter in everyday life.

Students can use addition and subtraction to determine how much an item costs and how much money they should receive back after making a purchase.

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3. Essential Question

How can I use math to make a purchase and determine how much change I should receive?

Supporting Questions

- How much does an item cost?
- How much money do I need to pay?
- How can I find the total cost of multiple items?
- How can I determine my change?
- How can I check whether my answer makes sense?

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4. Learning Objectives

Primary Objective

Given a simulated shopping situation, price tags, play money, and access to a calculator when appropriate, students will calculate the total cost of one or more items and determine the correct amount of change with **at least 80% accuracy on individualized tasks**.

Student-Friendly Objective

“I can use money to buy something and figure out my change.”

Additional Objectives

By the end of the lesson, students will be able to:

1. Identify the monetary value of common bills and coins.
2. Identify the price of an item.
3. Add the prices of two or more items when appropriate.
4. Identify an appropriate amount of money to use for payment.
5. Calculate the amount of change received.
6. Use a calculator or other mathematical tool appropriately when provided as an accommodation.
7. Explain or demonstrate the steps used to solve a money problem.

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5. Success Criteria

Students will demonstrate mastery when they can complete the assigned task with **80% accuracy**.

Students will know they are successful when they can:

1. Look at the price.
2. Add the prices if needed.
3. Decide how much to pay.
4. Subtract to find the change.
5. Check the answer.

Visual:

LOOK → ADD → PAY → SUBTRACT → CHECK

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6. Prior Knowledge

Students should have some familiarity with:

- Identifying common coins and bills
- Counting money
- Basic addition and subtraction
- Reading numbers

- Using a calculator, when appropriate
- Following a sequence of steps

Because students in a Life Skills classroom may have significantly different instructional levels, prior knowledge should be determined through teacher observation and student IEP goals.

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7. Materials

- Play coins and bills
- Price tags
- Small classroom “store”
- Shopping baskets or bags
- Pencils
- Paper
- Individualized shopping worksheets
- Calculators
- Number lines
- Hundreds/tens/ones or place-value supports, if needed
- Visual step-by-step procedure
- Whiteboard or interactive board
- Teacher observation checklist
- Exit ticket
- Optional real-world items such as pencils, notebooks, folders, snacks, water bottles, or school supplies

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8. Vocabulary

Vocabulary	Student-Friendly Definition
Price	How much something costs
Cost	The amount of money needed to buy something
Total	The amount altogether
Pay	Give money for something
Change	Money given back after you pay
Dollar	A unit of money worth 100 cents
Cents	Smaller units of money

Vocabulary	Student-Friendly Definition
Budget	A plan for how much money you can spend

9. Universal Design for Learning (UDL)

Multiple Means of Representation

Information will be presented in several ways:

- Teacher explanation
- Written examples
- Visual step-by-step directions
- Demonstration with play money
- Price tags
- Number lines
- Worked examples
- Verbal repetition of important directions

Multiple Means of Engagement

Students will:

- Participate in a simulated classroom store.
- Select items they would realistically purchase.
- Manipulate physical money.
- Work individually and/or with a partner.
- Receive immediate feedback.
- Choose between different items when appropriate.

The lesson will connect mathematics to a functional situation students may encounter outside of school.

Multiple Means of Action and Expression

Students may demonstrate understanding by:

- Writing an answer
- Selecting an answer
- Manipulating play money
- Using a calculator
- Pointing to the correct amount
- Verbally explaining their answer

- Demonstrating a purchase
- Completing a teacher-directed transaction

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10. Accommodations

Accommodations should be based on individual student IEPs. Potential accommodations may include:

- Extended time
- Calculator
- Reduced number of problems
- Directions read aloud
- Directions repeated or rephrased
- Visual directions
- Step-by-step checklist
- Number line
- Manipulatives
- Frequent teacher check-ins
- Small-group instruction
- Preferential seating
- Reduced visual distractions
- Verbal prompting
- Modeling before independent work
- Allowing students to respond verbally or nonverbally
- Breaking assignments into smaller sections

Important: Only accommodations listed in a student's IEP should be used as formal IEP accommodations.

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11. Differentiation

Because a high school Life Skills classroom may contain students working at different instructional levels, the same overall competency will be addressed through different levels of complexity.

Level 1 – Foundational

Students will:

- Identify coins and bills.

- Match money to a price.
- Identify which amount is greater.
- Practice paying for one item using exact change.

Example:

A pencil costs $1.

Which can you use?

$1 bill → Correct

Level 2 – Developing

Students will:

- Add the cost of two items.
- Determine change from $5 or $10.
- Use a visual procedure.

Example:

Notebook = $3
Pencil = $1

Total = $4

Pay with $5.

Change = $1.

Level 3 – More Independent

Students will:

- Add multiple items.
- Calculate change from larger amounts.
- Work with dollars and cents.
- Use a calculator independently when appropriate.
- Solve short shopping word problems.

Example:

Notebook = $3.50
Folder = $2.50

Total = $6.00

Pay with $10.

Change = $4.00.

Extension

Students will work with a budget.

Example:

You have $10.

You want to buy:

- Notebook: $3
- Pencil: $1
- Folder: $2

How much will you spend?

How much money will you have left?

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12. Teacher Preparation

Before students enter the classroom, the teacher will:

1. Set up a small classroom store.
2. Place price tags on items.
3. Organize play money.
4. Prepare individualized worksheets.
5. Place calculators and visual supports where students can access them.
6. Display the lesson objective.
7. Display the visual procedure:

LOOK → ADD → PAY → SUBTRACT → CHECK

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13. Minute-by-Minute Procedure

0–5 Minutes — Opening/Warm-Up

Teacher Actions

The teacher will greet students and introduce the lesson.

Teacher Script:

“Today we are going to practice something that we use in everyday life: buying things. We are going to use math to figure out how much something costs and how much change we should get back.”

The teacher will display several bills and coins.

“Who can tell me what this is?”

“How much is this worth?”

“Which is worth more: $5 or $10?”

The teacher will briefly review money identification as needed.

Student Actions

Students will:

- Identify bills and coins.
- State or indicate monetary values.
- Compare amounts.
- Respond verbally, with gestures, or by pointing.

Formative Assessment

Teacher will observe whether students can identify common monetary values.

5–10 Minutes — Introduce the Objective and Vocabulary

Teacher Actions

The teacher will display:

I CAN USE MONEY TO MAKE A PURCHASE AND FIND MY CHANGE.

Teacher:

“Today we are going to learn five steps.”

Point to each visual:

LOOK → ADD → PAY → SUBTRACT → CHECK

Teacher:

“First, we look at the price.”

“Second, we add the prices if we are buying more than one thing.”

“Third, we decide how much money we are going to pay.”

“Fourth, we subtract to find our change.”

“Finally, we check our answer.”

Student Actions

Students will repeat the steps or demonstrate them using the visual.

10–20 Minutes — Explicit Teacher Modeling

Example 1

Teacher displays:

Pencil = $1

Teacher:

“The pencil costs $1. If I give the cashier $1, how much change should I get?”

Students respond.

Teacher demonstrates using play money.

$1 − $1 = $0

Teacher:

“I paid exactly the amount it costs, so I get $0 change.”

Example 2

Teacher displays:

Notebook = $3

“You have $5. The notebook costs $3. How can we find the change?”

Teacher models:

$5 − $3 = $2

Teacher physically removes $3 from $5 using play money.

“So if I pay $5 for something that costs $3, I should get $2 back.”

Example 3

Teacher displays:

Pencil = $1
Notebook = $3

Teacher:

“What if I want both?”

Model:

$1 + $3 = $4

“If I pay with $5:”

$5 − $4 = $1

Teacher:

“My total is $4, and I should receive $1 in change.”

20–30 Minutes — Guided Practice

The teacher will establish a classroom store.

Possible items:

- Pencil — $1
- Folder — $2
- Notebook — $3
- Water — $1.50
- School supplies — $2.50

Students will work through problems with teacher guidance.

Problem 1

“Choose a pencil. How much does it cost?”

Problem 2

“Choose a notebook. If you pay with $5, how much change do you get?”

Problem 3

“Choose a pencil and folder. What is the total?”

The teacher will provide prompting as needed.

Teacher Prompts

Instead of immediately providing the answer, the teacher will use prompts such as:

- “What does the price tag say?”
- “What is the first step?”
- “Do we need to add?”
- “How much are you paying?”
- “What operation should we use?”
- “Does your answer make sense?”

30–42 Minutes — Individualized Student Practice

Students will complete tasks matched to their instructional level.

Level 1

Students will identify money and match money to prices.

Level 2

Students will solve one- and two-item shopping problems.

Level 3

Students will solve multiple-item shopping problems involving dollars and cents.

The teacher will circulate and collect formative assessment data.

Students will be encouraged to use their available supports rather than having the teacher provide the answer.

42–50 Minutes — Simulated Shopping Activity

Students will complete a simulated purchase.

Each student will:

1. Select an item.
2. Read or identify the price.
3. Select the money needed to pay.
4. Give the money to the teacher acting as the cashier.
5. Calculate the expected change.
6. Receive play money as change.
7. Check the amount received.

Teacher Script

“Welcome to our store! What would you like to buy?”

“How much does it cost?”

“How much are you paying?”

“How much change should you get?”

“Let’s check.”

The teacher will compare the student’s calculated answer with the actual change.

50–55 Minutes — Closure and Exit Ticket

The teacher will bring students back together.

Teacher:

“Today we practiced using money. What are our five steps?”

Students will identify:

LOOK → ADD → PAY → SUBTRACT → CHECK

Exit Ticket

Students will complete one individualized problem.

Example:

A notebook costs $3.

You pay with $5.

How much change should you receive?

Answer: $2

Students may demonstrate the answer through writing, selecting an answer, manipulating money, using a calculator, or verbally responding according to their individual needs.

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14. Formative Assessment

Formative assessment will occur throughout the lesson through:

- Observation
- Questioning
- Student responses
- Manipulation of play money
- Guided practice
- Teacher prompting
- Classroom-store transactions
- Individual practice

Teacher Observation Checklist

Skill	Independent	With Prompt	Not Yet
Identifies price	☐	☐	☐
Identifies money value	☐	☐	☐
Adds item prices	☐	☐	☐
Selects payment	☐	☐	☐
Calculates change	☐	☐	☐
Uses calculator appropriately	☐	☐	☐
Checks answer	☐	☐	☐

15. Summative Assessment

Students will complete a brief individualized assessment containing **five money problems**.

The assessment will be differentiated according to each student's instructional level.

Mastery Criterion

Students will demonstrate mastery by correctly completing **at least 4 out of 5 assigned problems (80%)**.

Students who do not meet the mastery criterion will receive additional instruction and practice using the visual supports and manipulatives before being reassessed.

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16. Assessment Examples

Foundational Assessment

1. Circle the $5 bill.
2. Match $1 to the correct price.
3. Select the money needed to buy an item costing $2.

Developing Assessment

1. Pencil = $1. You pay $5. What is your change?
2. Folder = $2 and pencil = $1. What is the total?
3. Notebook = $3. You pay $5. What is your change?

More Independent Assessment

1. Notebook = $3.50 and folder = $2.50. What is the total?
2. You pay $10. How much change should you receive?
3. You have $15. You spend $8. How much money remains?

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17. Teacher Reflection

Following the lesson, the student teacher will reflect on the following questions:

1. Did students demonstrate understanding of the lesson objective?
2. Which students met the 80% mastery criterion?
3. Which students required prompting?
4. Which accommodations were effective?
5. Were the visual supports effective?
6. Did students use the play money appropriately?
7. Was the level of difficulty appropriate for each student?

8. What misconceptions did students demonstrate?
9. What should be retaught?
10. What should be changed before the next lesson?

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18. Possible Follow-Up Lessons

This lesson could serve as the first lesson in a larger functional mathematics unit.

Lesson 2

Creating and Following a Budget

Students will receive a specific amount of money and determine which items they can afford.

Lesson 3

Comparing Prices

Students will compare prices at two stores and determine which option costs less.

Lesson 4

Making a Shopping List

Students will create a shopping list and calculate an estimated total.

Lesson 5

Independent Shopping Simulation

Students will independently complete a simulated shopping experience using a budget.

Lesson 6

Real-World Application

If appropriate within the school's instructional program, students can apply the skills to a supervised school/community-based purchasing activity.

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19. Rationale

This lesson was designed to connect mathematical instruction to functional skills that students may use in everyday life. Rather than teaching addition and

subtraction only through abstract worksheets, students will apply those skills to a familiar situation involving purchasing items and receiving change.

The lesson also allows for significant differentiation without changing the overall competency. Students working on foundational skills can identify money and match prices, while students with stronger mathematical skills can calculate totals, change, and budgets. This allows students with different instructional levels to participate in the same classroom activity while working toward appropriately challenging objectives.

The use of visual supports, concrete manipulatives, modeling, repetition, calculators, and multiple response options provides students with several ways to access and demonstrate mathematical understanding. This is particularly appropriate for a Life Skills classroom because the goal is not only mathematical accuracy but also the development of functional independence.

Pennsylvania's mathematics framework emphasizes both procedural skills and conceptual understanding and highlights the application of mathematics to real-world challenges. Pennsylvania's current Personal Finance standards also emphasize spending and financial decision-making, making a functional money lesson particularly relevant for high school students.

Student Teacher Notes

Before teaching: Review the cooperating teacher's current curriculum, students' IEP goals, accommodations, and present levels of performance. Adjust the differentiated tasks accordingly.

During teaching: Focus on prompting students to use the visual sequence rather than immediately providing answers.

After teaching: Record which students were independent, which required prompting, and which skills should be addressed in the next lesson.